

therapy

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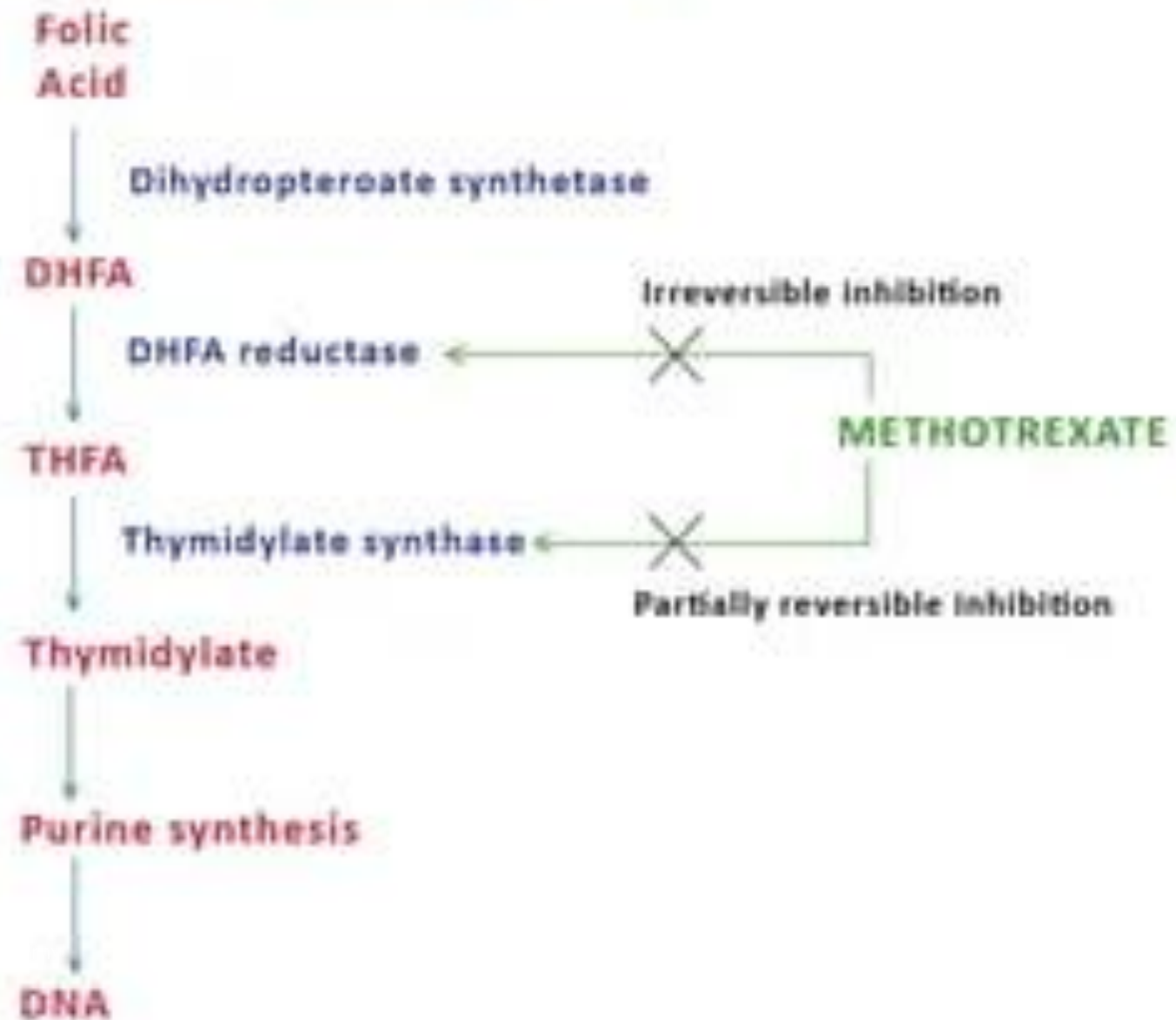
SCE Dermatology / EBDV by dr :Eman Gomaa - CoNNect Academy

Metotrexate



Pharmacokinetics	Absorption :oral , parenteral Metabolism :liver Excretion :kidney
Mechanism	Prodrug → methotrexate polyglutamate Competitive irreversible inhibition of DHFR
Action	Antiinflammatory Antiproliferative Immunosuppressive

MECHANISM OF ACTION



methotrexate

Formulation	Tab 2,5 mg Vial 50mg /2m
Indication	FDA : ps . Sezary syndrome Offlabel :prp, pleva , plc , mf
Dose	15-25 mg /week Once weekly with folic acid

Oral (2.5 mg tablets)	Parenteral (50 mg vial 1 ml/2 ml)
15 mg / week Low-dose MTX مهم	25 mg / week IM ½ الحقنه عضل كل اسبوع
Single dose OR 3 divided doses within 12-hr interval i.e. 2 tablets (5 mg) x 3 times [24 hrs]	
Folic acid supplementation: 1-2 tablets daily <u>EXCEPT</u> day(s) of MTX	

Methotrexate

Side effect	(bone , marrow , liver) Mild common : neuse , vomiting Sver : Bone marrow suppression Hepatotoxicity Pulmonary toxicity Mtx toxicity Teratogenicity
Monitoring	CBC , LIVER , CHEST XRAY
Drug interaction	Hepatotoxicity :retenoid , alcoh Pancytopenia :sulpha , dapson , phenytoin Category x

	Baseline	Cumulative dose
Low-risk	Not necessary	3.5 – 4 grams MCQ
High-risk	Baseline, or after 2-6 month of therapy	1 – 1.5 grams

Systemic retinoids

1 st generation	Tretenoin , alitretenoin
2 nd generation	Etretine , acitretine ,
3 rd generation	Bexaroten , adapalin , tazarotene

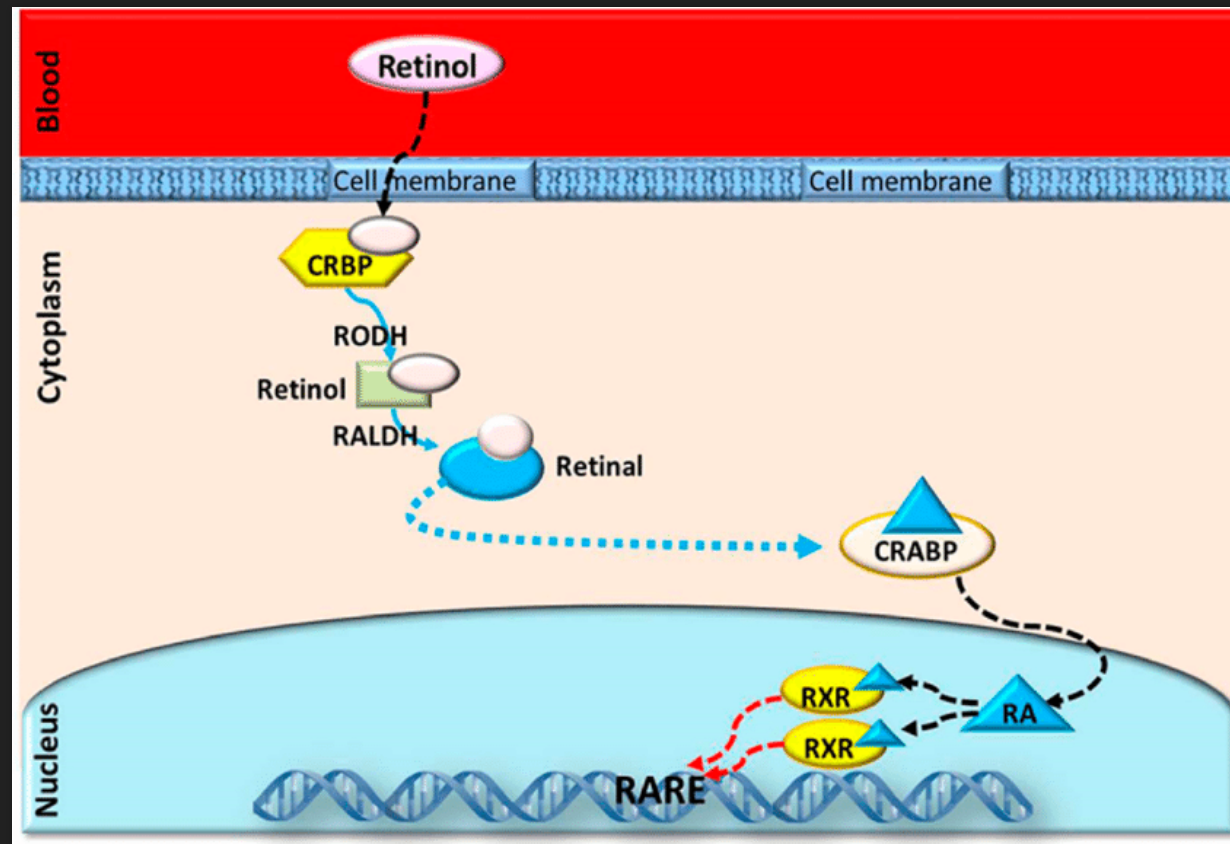


Systemic retinoids

	Absorption :Fatty meals Metabolism :hepatic Excretion :biliary , renal
Mechanism	CRAB In nucleus :2 receptor RAR: tretinoin , alitretinoin RxR :bexarotene, alitretinoin
Action	Antiproliferative Antiinflammatory Antiseborrheic Tumor preventive therapy
Formulation	Isotretinoin :10 , 20 , 40 Neotigazone 10 , 25

Systemic retenoids

Indication	FDA iso:acne Acitretine :ps Bexa : mf Alitrtinoin : hand eczema Offlabel :
Dose	,5 -1 mg /kg Cumulative dose 120mg/kg/course
Side effect	Mucocutaneus , ocular dryness Teratogenicity Hypertriglycerdemia Hepatic Neurological :pseudotumor cerebri Bone :premature epiphysial closure Muscle pain Leukopenia Hypothyroidism



	Retenoids
Interaction	Doxycyclin Alcohol Methotrexat Vit A
	Category x

Cyclosporine



Chemistry	Soil fungus
Absorbtion	Empty stomch
Metabolizim	Ccyp 450 , cyp 3A4 no grap fruit Excretion :bile , feces
Action	Immunosuppresion
Mechanisim	Systemic calcinurine inhibitor → decrease T cell activation
Formulation	Sandimmune , neoral 25, 50 , 100 mg cap

Indication	FDA: ps , atopic dermatitis
Dose	2,5 -5 mg /kg /day
Side effect	• Nephrotoxicity - Vc→hypoxia → free radicle →revesable • Hypotension • Maligancy • Hypertrichosis , hyperpigmentation • Hyperlipidemia , hyperuricemia • Hyperkalemia • Hypomagnesemia

Monitoring	ABP , serum creatinine , urin analysis
Interaction	Renal toxicity :NSAID , aminglycosid
	Category c No children <16 years

	METHOTREXAT	ACETRITINE	CYCLOSPORINE
DOSE	Single oral/IM weekly dose (7.5 - 25 mg) + oral folic acid 1-5 mg daily except for the day of methotrexate	10-50 mg/day given as a single dose for 3 - 6 months	2.5-5 mg/kg/day in 2 divided doses (for maximum 1 - 2 years)
S . E	Nausea / vomiting, Hepatotoxic, Bone marrow suppression, immuno-suppression, teratogenic and pulmonary fibrosis	Teratogenic Hypelipidemia Cheilitis/Xerosis Hepatotoxic	Nephrotoxic, Hypertension, Malignancy, immune-suppression, Hypertrichosis, Gingival hyperplasia
contraindication	Pregnancy (X) Lactation Liver/kidney disease Anaemia, leucopenia or thrombocytopenia Active infection Hypersensitivity to	Pregnancy (X) Lactation Hypelipidemia	Abnormal renal function Uncontrolled hypertension Malignancy Hypersensitivity to Cyclosporine Live vaccinations or

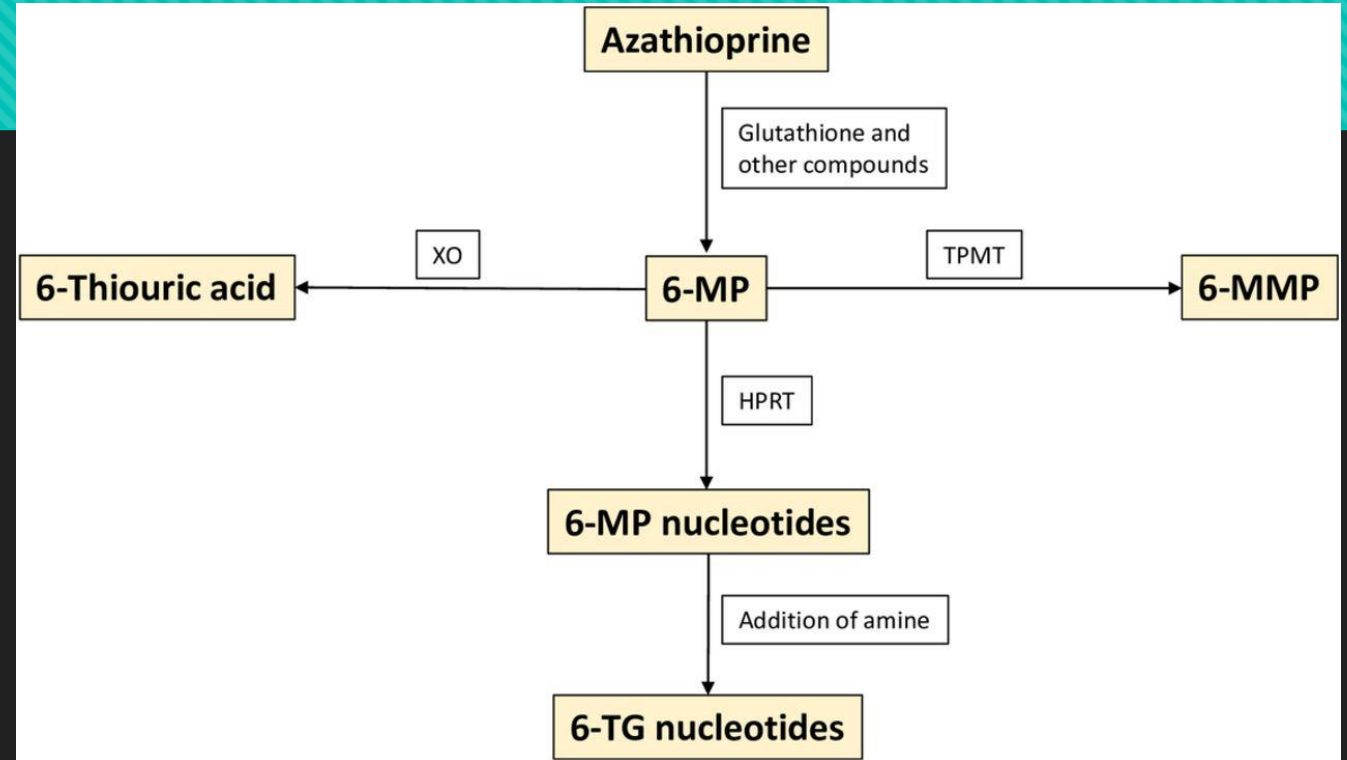
monitoring	CBC Liver/kidney function tests Pregnancy test Tests for HIV/PPD Chest x-ray Liver biopsy	Pregnancy test Lipid profile CBC Liver/kidney function tests	Blood pressure Kidney function tests (BUN, creatinine and Urine analysis) Lipid profile

Azathioprine

	Synthetic purin analogue , well absorbed , 88%
Pharmacokinetic	Well absorbed Metabolism : AZA prodrug → 6 mercaptopurine 1. HGPRT → active 6 thioguanine 2. TPMT → inactive 3. XANTHINE OXIDASE → inactive
Mechanism	Decrease purine metabolism → antiproliferative → on lymphocyte Decrease T cell , B cell Decrease : Langerhans cells

Azathioprine

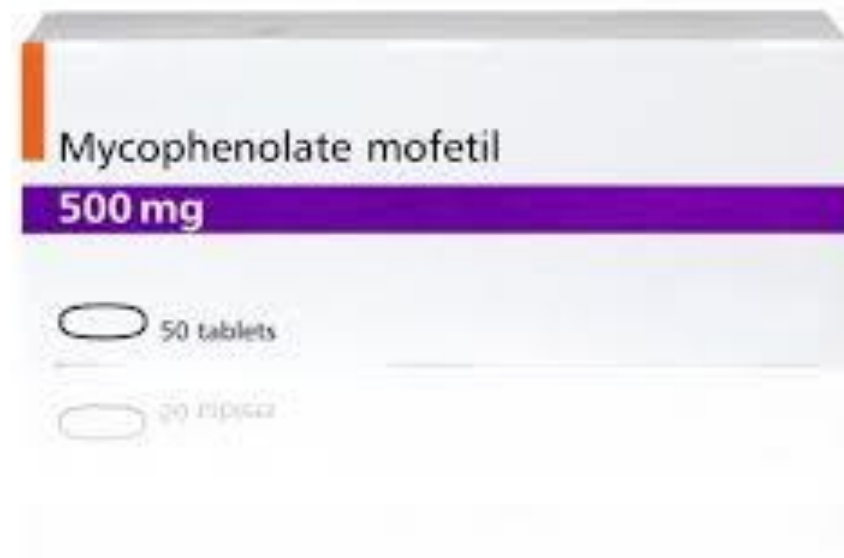
Indication	No FDA approved in dermatology Steroid sparing agent
Dose	1-3 mg /kg /day take with food
Side effect	Neusea , vomiting , myelosupp Lymphoma , scc Hypersensitivity
Monitoring	Tpmt , cbc Liver , kidney ft
	Category D



Mycophenolate mofetil MMF

	Prodrug MMF → mycophenolic acid Absorption : rapid absorption
Mechanism	MPA → decrease IMPDH → decrease de novo pathway of purin biosynthesis Decrease B , T lymphocyte only
Action	Immunosuppression
	250 – 500 mg cap myfortic
	FDA : in renal , cardiac , liver allograft rejection prevention
SE	V mild Diarrhea → most common

Monitoring	Cbc , plt count , liver , kidney
	Category D



THALIDOMIDE

	NON Barbiturate , sedative , hypnotic
	Anti inflammatory – immunomodulatory Anticancer effect
Mechanism	Unkown Decrease tnf alpha Decrease il6, il12 Increase apoptotic pathway
Formulation	Thalix 50 -100- 150 -200 cap
Indication	FDA erythema nodosum leprosum . MM Off label : prurigog nodularis , mucosal ulceration

Dose	50-150 mg / day
S.E	Teratogenic : singl dose → phacomelia Sedation Peripheral neuropathy Thrombosis
Monitoring	Pregnancy category x





Antimalarial



	Well absorbed
Action	Antiinflam- ,immunosuppression , photoprotective
Mechanism	Stabilization of lysosome Decrease antigen presentation Decrease TLR , CMI Photo sink
Indication	FDA :cutaneous LE Off label :photosensitive dermatosis granulomatous dermatosis painniculitis

Dose	200 mg plaquanil -250 mg chloroquine
Side effect	Retinopathy , blindness Hemolysis pigmentation Depigmentation of hair Worse ps , porphyria
Monitoring	Ocular , CBC , G6PD
Interaction	Smoking decrease efficacy
	Chloroquin →D , hydroxychloroquine →C

DAPSONE

di amino di phenyl sulphone

Action	Antineutrophilic , antimicrobial , anti-inflammatory
	Antifolat
Formulation	50 -100 mg tab
Indication	FDA :leprosy , DH Off label : neutrophilc dermatosis
Dose	Leprosy 100 mg tab Dermatological :50 mg /day
S.E	Hemolysis , hemolytic anemia Methemoglobinamia :ferrous to ferric Leukopenia agranulocytosis

Dapson

Monitoring	Cbc , G6PD , LFT , KFT
INTERACTION	Methitrexat , sulpha →anti folat Antimalrial →hemolysis
	Category c
	Safe in children



Name	Mechanism of action	Spectrum, category	Miscellaneous
Acyclovir	Phosphorylated by viral thymidine kinase to acyclovir monophosphate, which blocks viral DNA polymerase → stops viral DNA synthesis	Herpes simplex virus (HSV), varicella-zoster virus (VZV) Pregnancy category B	<u>SE</u> : IV infusion associated with reversible obstructive nephropathy, rarely may see severe CNS changes (ie. seizures)
Valacyclovir	Prodrug of acyclovir, same mechanism of action (viral thymidine kinase-dependent activity)	HSV, VZV, cytomegalovirus (CMV) Category B	Better bioavailability than acyclovir <u>SE</u> : TTP/HUS* seen in advanced HIV disease and transplant patients taking high doses
Penciclovir	Phosphorylated by viral thymidine kinase (similar mechanism to acyclovir)	HSV, VZV	Low bioavailability so typically used in topical form
Famciclovir	Prodrug of penciclovir with same mechanism as above	HSV, VZV Category B	Better bioavailability than penciclovir
Ganciclovir	Phosphorylated by viral thymidine kinase; same mechanism as above	CMV (retinitis and CMV prophylaxis in transplant pts) Category C	Better activity against CMV than acyclovir; ↓ oral bioavailability <u>SE</u> : neutropenia, bone marrow suppression, mucositis, thrombocytopenia, seizures hepatic dysfunction
Foscarnet	Noncompetitive inhibition of viral DNA polymerases; analogue of pyrophosphate Does not require phosphorylation so active against acyclovir-resistant viruses	CMV (retinitis), resistant HSV, resistant VZV Category C	Only IV form; active against infections resistant to acyclovir, famciclovir, ganciclovir <u>SE</u> : penile ulcerations or erosions , nephrotoxicity
Cidofovir	Nucleoside analogue, inhibits viral DNA polymerase, independent of thymidine kinase activation	CMV Category C	IV only; active against infections resistant to ganciclovir/foscarnet <u>SE</u> : renal proteinuria, renal toxicity, ↑ creatinine
Amantadine, Rimantadine	Inhibit uncoating of viral DNA	Influenza A/C, rubella	<u>SE</u> : anticholinergic symptoms,

